

Information Retrieval -Introduction-

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Riwayat Pendidikan :

- SD Kanisius Kobong II Semarang, lulus tahun 1983
- SMP Kanisius Raden Patah Semarang, lulus tahun 1986
- SMA Negeri 3 Semarang, lulus tahun 1989
- S1 Matematika UNDIP Semarang, lulus tahun 1996
- S2 Teknik Informatika STTIBI Jakarta, lulus tahun 2001
- Internship Program Doktorat Speech Processing- Tokyo University of Technology (TUT), Japan, lulus tahun 2014
- S3 Teknik Elektro, ITS Surabaya, lulus tahun 2016

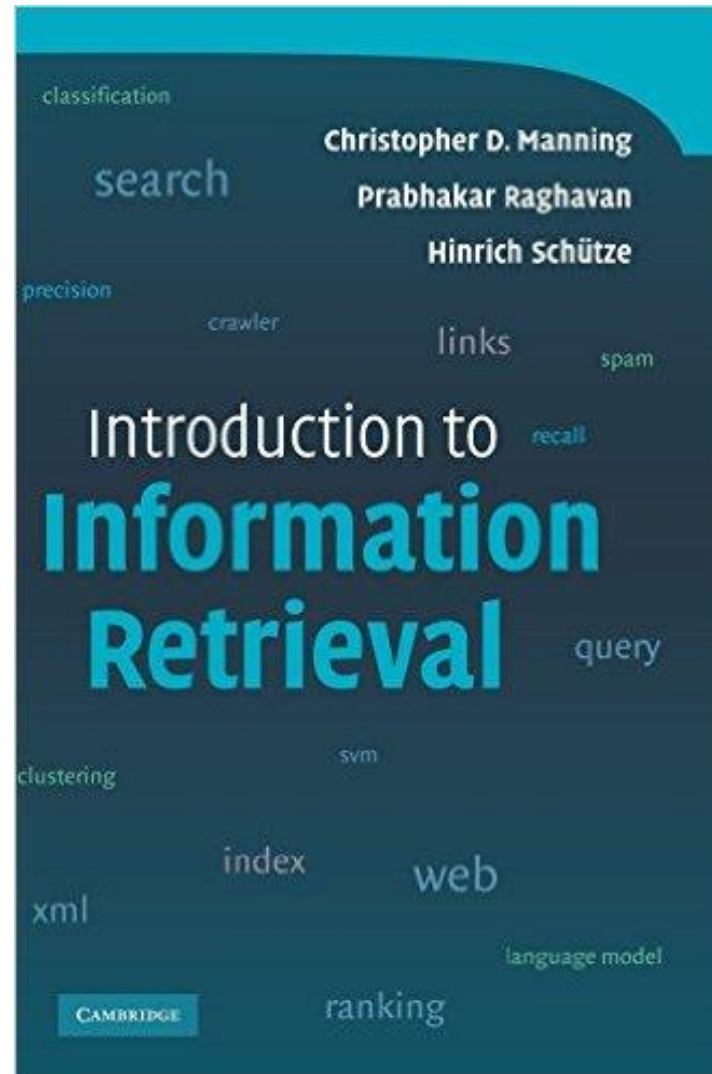
Pengalaman Tambahan :

- Workshop Datamining, PENS ITS Surabaya, 2006
- Soft Computing and Pattern Recognition (SOCPAR), Universiti of Teknikal Malaysia Melaka (UTeM), Malaysia, 2009.
- Seminar dan Workshop Soft Computing, South China University of Technology (SCUT), Guangzhou, China, 2013
- Visiting Professor (Dosen Tamu) di University of Split Croatia, Program Beasiswa Erasmus⁺ Uni Eropa, 2018
- Workshop Big Data Analysis dan IOT, Big Data Alibaba, Fozhou Polytechnic, Fuzhou China, November sd Desember 2019

Topics

1. Introduction
2. IR Model
3. Document Classification: k-NN and Naive Bayes
4. Document Clustering: K-Means
5. Document Clustering: Hierarchical Agglomerative Clustering (HAC)
6. Document Clustering: Self Organizing Map (SOM)
7. Features Reduction
8. Document Summarization
9. Rapidminer
10. Weka
11. Lucene
12. Python

Handbook



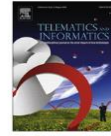
Paper

Proceeding

- Annually
- 2-6 pages
- IEEE, presentation is compulsory

Journal

- 2 times or more publication every year
- More than 6 pages



Clustering results of image searches by annotations and visual features



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ABSTRACT

Image on web has become one of the most important information for browsers; however, the large number of results retrieved from images search engine increases the difficulty in finding the intended images. Image search result clustering (ISRC) is a solution to this problem. Currently, the ISRC-based methods separately utilized textual and visual features to present clustering result. In this paper, we proposed a new ISRC method as called Incremental-Annotations-based image search with clustering (IAISC), which adopted annotation as textual features and category model as visual features. IAISC can provide clustering result based on the semantic meaning and visual trail; further, presented by the iteratively structure, a user can obtain the intended image easily. The experimental result shows our method has high precision that the average precision rate is 73.4%; particularly, the precision rate is 96.5% when the user drills down the intended images till the last round. Regarding efficiency, our system is one and a half times as efficient as the previous studies.

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1. Introduction

Image on web has become one of the most important information for users these days. Since there are a huge number of images on the web, it is hard to find the intended images by simple query. The results of a query may contain several topics; for example, a query “Pluto” in Google Images, the results contain two different scope of semantic meaning: one is Pluto in the solar system; the other is the dog named “Pluto” in Disney world (Cai et al., 2004). It is worth noting that the results of the two scopes are mingled, that means it is hard to identify the required images immediately. Since the search engine does not clustering the results beforehand, the users have to browse all the images through page by page, and then they should manually organize these images.

Recently, web image search engine, like Google image, only returns a small fixed number of images (Wang et al., 2007). In the perspective of users, it becomes even harder to find the intended images when different scopes mingled together. To solve the problem, technique of image search result clustering (ISRC) was proposed (Cai et al., 2004; Wang et al., 2007; Ding et al., 2008; Jing et al., 2006) which is clustering images based on visual or textual features.

Regarding the visual features, such as color or texture, can be used to find similar images (Cai et al., 2004); however, the features have some limitations that it is hard to present the semantic meaning (called semantic gap) with the visual features and it takes too much time to extract the visual features from high-definitional and high-dimensional images.

Regarding the text features, it means the features of the text surrounding the image or other text sources. Text-based ISRC (Wang et al., 2007; Ding et al., 2008; Jing et al., 2006) extracted key phrases from the texts to clustering images into several clusters in a reasonable time, compared to the time to clustering images by visual features. With the text features, images in each cluster then have semantic meaning and can be used for further application. Although the images are annotated with

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Grouping WWW Image Search Results by Novel Inhomogeneous Clustering Method

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Abstract

In this paper, a novel inhomogeneous clustering method is proposed for grouping web images. It is used to re-organize the search result of web image search engines into a hierarchical structure so that the users can conveniently browse the search result. This method takes into account various features associated with web images, and treats them in different ways. For the surrounding text extracted from the containing web pages, co-clustering approach is adopted; for low-level features of the image content and other features, one-way clustering approach is adopted. The clustering results of different approaches are combined together to produce the final image groups. Experimental results demonstrate the effectiveness of the proposed method.

1. Introduction

WWW image search engines [1] are powerful tools to search for digital images on the Internet by keywords. Unlike traditional image databases with manually labeled annotations, web image search engines index images with some text-based features, such as image file names or the surrounding text in containing web pages. Those features may be regarded as an approximate description of the image content. When the user submits a keyword query, the system typically produces a ranked list of images according to the relevance of image's text description to the user's query.

However, because of the ambiguities of keywords, the results of the existing search engines are still not satisfactory in many cases. Even if all images returned are relevant to the input keywords, it is yet difficult for the user to find the right images with his/her intended concepts or visual styles. Obviously, even if different users use the same keywords to search images, their objectives may be different. The one-dimensional search results produced by current search engines can not meet requirements of different users. Therefore, it will be

quite useful if we can automatically group search results into different clusters in terms of concepts and visual styles. In this manner, users are allowed to view the search results through a few clusters rather than jumbled images. Some studies also show that grouping images by visual features can help the user browse search results [2].

As web images are indexed with text information, the co-clustering method [5] used to cluster both terms and documents may be adopted here. However, as the surrounding text automatically extracted from containing web pages are not accurate enough, other information of images such as low-level features and hyperlink structures, should be taken in account as well. But those features are quite different in nature with each other: discrete or continuous, dense or sparse, high-dimensional or low-dimensional. How to use them simultaneously is a challenging problem.

In this paper, we propose a novel parallel, hybrid clustering algorithm to process inhomogeneous information naturally. Every feature can select its “favorite” clustering algorithm, and its “contribution” can be merged into a global loss function. Using this algorithm, we can cluster search results of our web image search engine by keywords, low-level features and hyperlink information, and got encouraging experimental results.

The paper is organized as follows. Firstly, in Section 2, we introduce the related works. The detailed explanation of the proposed algorithm is presented in Section 3, including the flowchart of our algorithm and the discussion of the convergence. The experimental results are given in Section 4. Concluding remarks appear in Section 5.

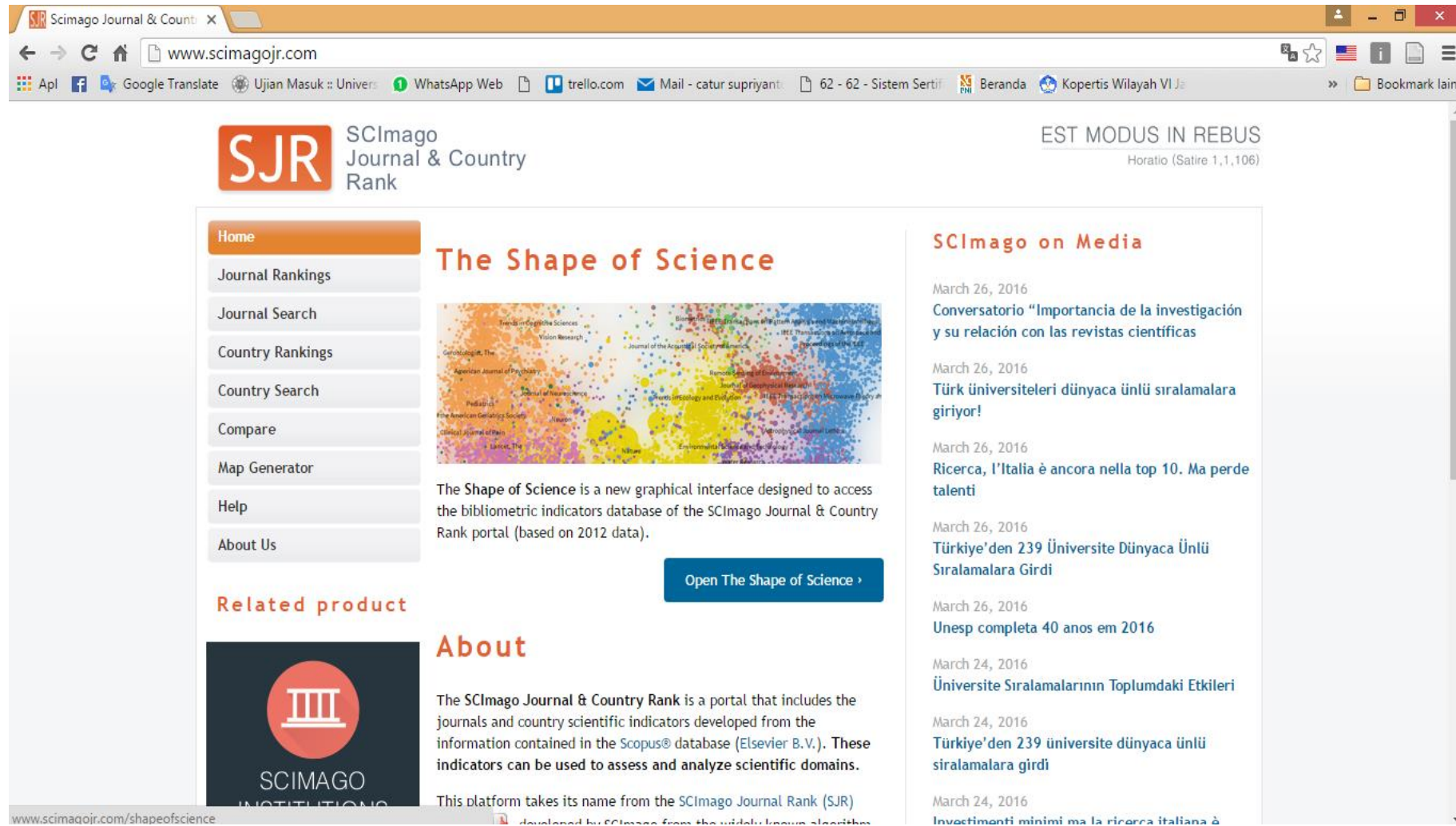
2. Related Work

In general, existing clustering algorithms may be classified into two types: one-side clustering and parallel clustering. The one-side clustering, also named one-way clustering, clusters along one dimension based on similarities with respect to other dimension (e.g. image

Indexing

1. International Scientific Indexing (ISI) <http://isindexing.com/>
2. Scopus scopus.com
3. Directory of Open Access Journals (DOAJ) doaj.org
4. Google Scholar

Scimago



Scimago (2)

The screenshot shows the Scimago Journal & Country Rank website. The browser address bar displays the URL: www.scimagojr.com/journalsearch.php?q=19500157821&tip=sid&clean=0. The website header includes the SJR logo, the text "SCImago Journal & Country Rank", and the motto "EST MODUS IN REBUS" with the quote "Horatio (Satire 1,1,106)".

On the left side, there is a navigation menu with the following links: Home, Journal Rankings, Journal Search (highlighted), Country Rankings, Country Search, Compare, Map Generator, Help, and About Us. Below the menu is a yellow button that says "Show this information in your own website".

The main content area is titled "Journal Search". It contains a search query input field, a dropdown menu set to "Journal Title", and a "Search" button. Below the search field is a checkbox labeled "Exact phrase".

The search results are for the "International Arab Journal of Information Technology". The results show the following details:

- Country: [Jordan](#)
- Subject Area: [Computer Science](#)
- Subject Category:

A table displays the journal's quartile performance over time. The table has columns for years from 1999 to 2014 and a row for the "Computer Science (miscellaneous)" category. The quartile values are as follows:

Category	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014
Computer Science (miscellaneous)											Q3	Q4	Q4	Q2	Q2	Q2

Below the table, the following information is provided:

- Publisher: [Zarqa University](#). Publication type: Journals. ISSN: 16833198
- Coverage: 2008-2015
- H Index: 10

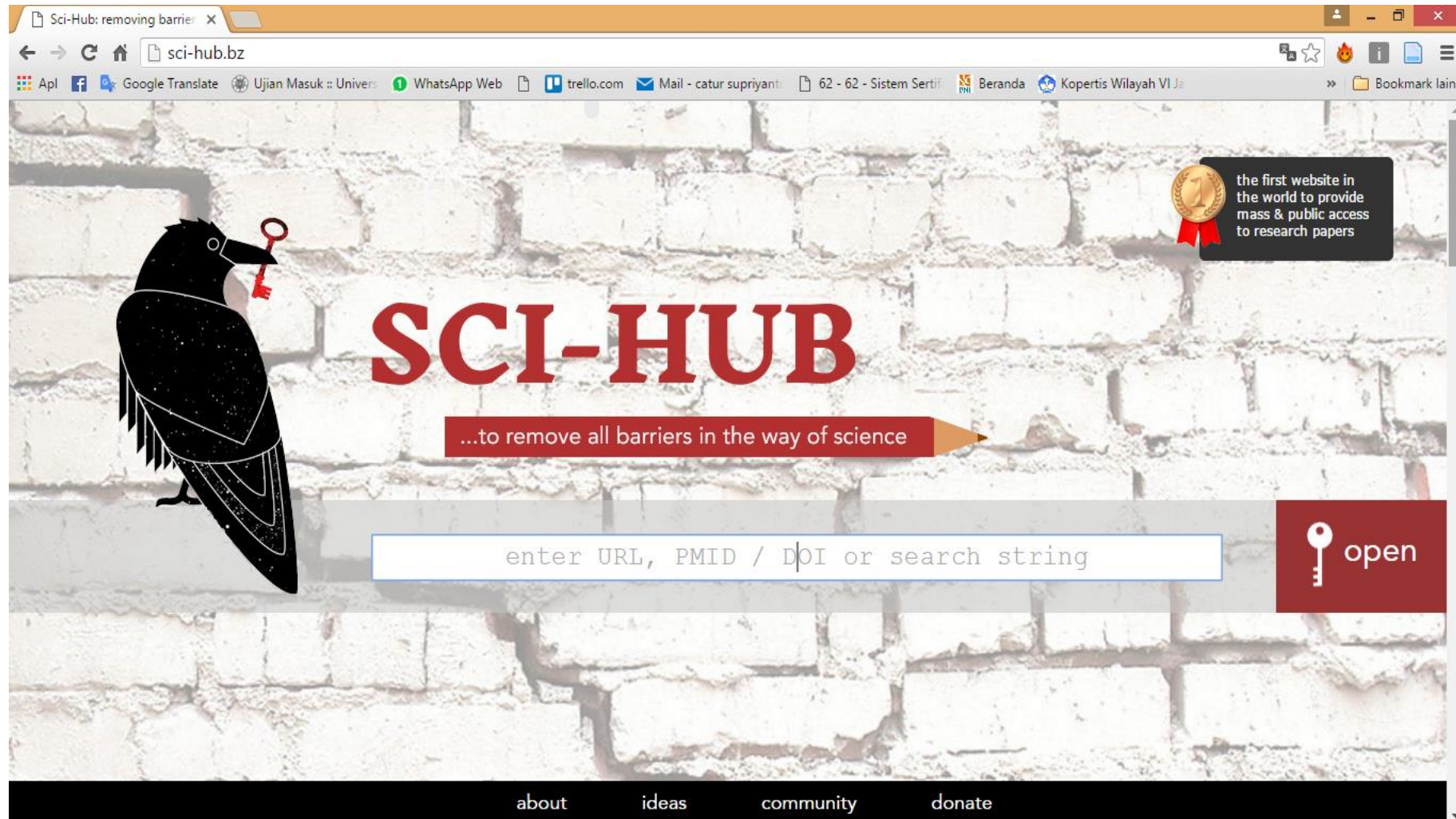
At the bottom right, there are buttons for "Charts" and "Data".

On the bottom left, there is a small box titled "International Arab Journal of Information Technology" containing a line graph showing the journal's performance over time. The graph has two indicators: SJR and Cites. The SJR value for 2007-2014 is 0.34, and the Cites value is 0.71.

Databases

1. ACM
2. Springer
3. Science Direct
4. IEEE
5. Google Scholar

Sci-Hub



Predator Journal

The screenshot shows a web browser window with the URL <https://scholarlyoa.com/individual-journals/>. The page has a blue header with the text "Scholarly Open Access" and "Critical analysis of scholarly open-access publishing". Below the header is a navigation bar with links: Home, About the Author, Disclaimer, LIST OF PUBLISHERS, and LIST OF STANDALONE JOURNALS. The "LIST OF STANDALONE JOURNALS" link is active. Below the navigation bar is a section titled "LIST OF STANDALONE JOURNALS" with the subtitle "Potential, possible, or probable predatory scholarly open-access journals". The text explains that this is a list of questionable, scholarly open-access standalone journals and provides instructions for how to use the list. It also includes a search bar and a list of recent posts. The recent posts list includes: "Three New Open-Access Publishers — All with Dumb Names", "Oncotarget's Peer Review is Highly Questionable", "Two Predatory Publishers: One Old, One New", "Two New Absurd OA Journals, Two New OA Mandates", and "Two Turkish Journals with a Fake, 'German' Editor". There is also an "ARCHIVES" section with a "Select Month" dropdown and a "CATEGORIES" section with links to "article processing charges", "Australia", and "Mandates".

LIST OF STANDALONE JOURNALS

Potential, possible, or probable predatory scholarly open-access journals

This is a list of questionable, scholarly open-access standalone journals. For journals published by a publisher, please look for the publisher on the list of publishers, [here](#). This list is only for single, standalone journals.

We recommend that scholars read the available reviews, assessments and descriptions provided here, and then decide for themselves whether they want to submit articles, serve as editors or on editorial boards. The criteria for determining predatory journals are [here](#).

We hope that tenure and promotion committees can also decide for themselves how importantly or not to rate articles published in these journals in the context of their own institutional standards and/or geo-cultural locus. We emphasize that journals change in their business and editorial practices over time. This list is kept up-to-date to the best extent possible but may not reflect sudden, unreported, or unknown enhancements

[Academic Exchange Quarterly](#)

[Academic Research Reviews](#)

[ntemporary Research Journal \(AOCRJ\)](#)

RECENT POSTS

- o Three New Open-Access Publishers — All with Dumb Names
- o Oncotarget's Peer Review is Highly Questionable
- o Two Predatory Publishers: One Old, One New
- o Two New Absurd OA Journals, Two New OA Mandates
- o Two Turkish Journals with a Fake, "German" Editor

ARCHIVES

Select Month

CATEGORIES

- o article processing charges
- o Australia
- o Mandates

Mendeley Desktop

The screenshot displays the Mendeley Desktop application window. The title bar reads "Mendeley Desktop". The menu bar includes "File", "Edit", "View", "Tools", and "Help". Below the menu bar is a toolbar with icons for "Add Files", "Folders", "Related", "Share", and "Sync". A search bar in the top right corner contains the text "bloom".

The left sidebar is divided into three sections: "Mendeley" (with "Literature Search" and "Mendeley Suggest"), "My Library" (with "All Documents", "Recently Added", "Recently Read", "Favorites", "Needs Review", "My Publications", "Unsorted", "Emotion Classification", "Plos One", "Porn Filter", "Systematic Literature Review", and "Create Folder..."), and "Groups" (with "Porn Filter" and "Create Group..."). At the bottom of the sidebar is the "Trash" section with "All Deleted Documents".

The main pane shows search results for "bloom" in "All Documents". The results are listed in a table with columns for star, document icon, title, authors, and abstract. The selected item is "How to Apply the Bloom Taxonomy to Software Engineering" by Motoei Azuma, F. Coalier, J. Garbajosa - 2003. The abstract states: "The BLOOM taxonomy is used in the SWEBOK to specify the expected level of understanding...".

The right pane shows the details of the selected document. It includes a warning message: "These details need reviewing. You can mark them as correct, or search the Mendeley catalog." with buttons "Details are Correct" and "Search". Below this, the "Type" is set to "Journal Article". The title is "How to Apply the Bloom Taxonomy to Software Engineering". The authors are "M. Azuma, F. Coalier, J. Garbajosa". There is a button "View research catalog entry for this paper". The "Journal:" field is empty, "Year:" is "2003", "Volume:" is empty, "Issue:" is empty, and "Pages:" is empty. The "Abstract:" section contains the text: "The BLOOM taxonomy is used in the SWEBOK to specify the expected level of understanding of each topic within its Knowledge Areas (KA) for a 'graduate plus four years of experience'. This paper discusses how Bloom's taxonomy could be expanded to be more useful not only for education but also for industry. A new taxonomy that is more applicable to engineeri...".

The status bar at the bottom indicates "1 of 1248 documents selected".

Zotero Desktop

Zotero

File Edit Tools Help

My Library

- 31 Oktober 2012
- Corpus
- Difabel
- diknas
- DKV
- ILKOM
- IndoSS
- Lip-reading
- POS Tagger
- Pustaka Paper - 1
- sail_publications_export_2013_09_14
- sail_publications_export_2013_09_15
- Sintesis Nada
- Speech Emotion
- Speech Synthesis
- VTTS
- My Publications
- Duplicate Items
- Unfiled Items
- Trash

Title	Creator
3D Lip-Synch Generation with Data-Faithful Machine Learning	Kim and Ko
3D models of the lips for realistic speech animation	Guiard-Marigny et al.
A Computer-animated Tutor for Spoken and Written Language ...	Massaro
A Framework of an Online Self-Based Learning for Teaching Ara...	Ibrahim et al.
A maximization technique occurring in the statistical analysis of ...	Baum et al.
A model for human faces that allows speech synchronized anim...	Parke
A Novel Visual Speech Representation and HMM Classification f...	Yu et al.
A Research Proposal to Address the Learning Strategies Used by...	Meurant
A Segment-based Audio-visual Speech Recognizer: Data Collec...	Hazen et al.
A synthetic speaker	Dudley et al.
A system for image-text relations in new (and old) media	Martinec and Salway
A virtual classroom platform based on text to visual speeches	Zhixiao et al.
An articulation model for audiovisual speech synthesis—Determ...	Fagel and Clemens
An articulation model for audiovisual speech synthesis—Determ...	Fagel and Clemens
An Artificial Talker Driven from a Phonetic Input	Jr and Gerstman
An Introduction to Text-to-Speech Synthesis	Dutoit
Animating speech: an automated approach using speech synthe...	Hill et al.
Application of Multimodal Learning in Online English Teaching	Sun
Applying a multimedia storytelling website in foreign language ...	Tsou et al.
Articulatory Features for Robust Visual Speech Recognition	Saenko et al.
Audio-visual and multimodal speech-based systems	Gibbon et al.
Audio-visual automatic speech recognition: An overview	Potamianos et al.
Audiovisual Speech Synthesis	Bailly et al.
Audiovisual speech synthesis: An overview of the state-of-the-art	Mattheyses and Verhelst
Automated lip-synch and speech synthesis for character animati...	Lewis and Parke
Automatic lip sync and its use in the new multimedia services fo...	Zorić and Pandžić
BRIX: meeting the requirements for online second language lear...	Sawatpanit et al.
Coarticulation in recent speech production models	Kent and Minifie
Compact 2D Facial Animation Based on Context-dependent Vis...	Costa and De Martino
Comparison of Phoneme and Viseme Based Acoustic Units for S...	Bozkurt et al.
Comprehensive many-to-many phoneme-to-viseme mapping a...	Mattheyses et al.
Computer-Assisted Audiovisual Language Learning	Wang et al.
Confusions Among Visually Perceived Consonants	Fisher

Info Notes Tags Related

Item Type Conference Paper

Title 3D Lip-Synch Generation with Data-Faithful Machine Learning

Author Kim, Ig-Jae

Author Ko, Hyeong-Seok

Abstract

Date 2007

Proceedings Title Computer Graphics Forum

Conference Name

Place

Publisher Wiley Online Library

Volume 26

Pages 295-301

Series

Language

DOI

ISBN

Short Title

URL <http://onlinelibrary.wiley.com/doi/10.1...>

Accessed 26/6/2016 23.03.32

Archive

Loc. in Archive

Library Catalog Google Scholar

Call Number

Rights

Extra

Date Added 26/6/2016 23.03.32

Modified 26/6/2016 23.03.32

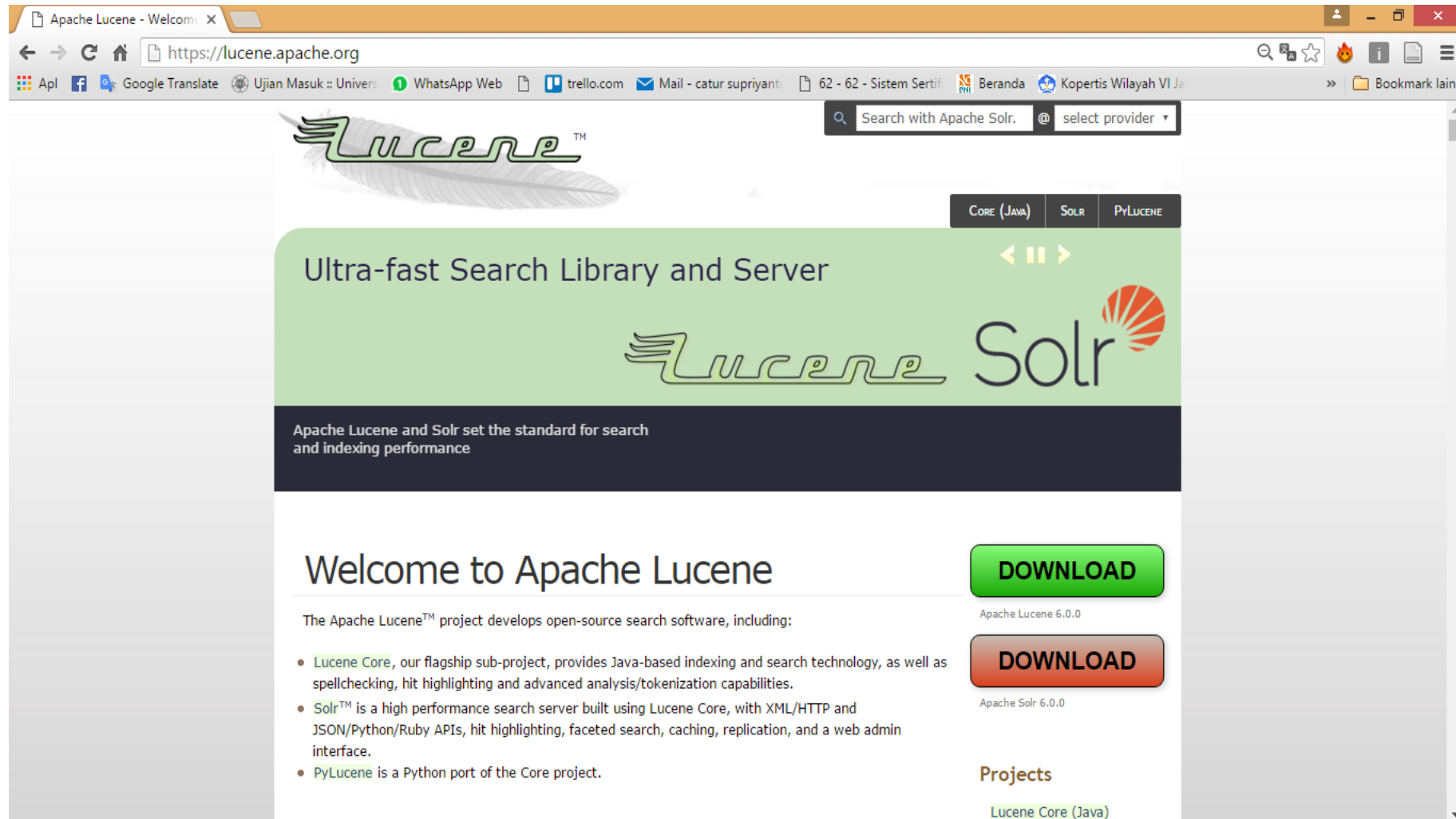
2D lip contour animation

3-tuple multimodal term

academic reading academic reaing

Access protocols Acoustic resonance

Lucene



The screenshot shows the Apache Lucene website in a web browser. The browser's address bar displays 'https://lucene.apache.org'. The website features a green header with the Lucene logo and a search bar. Below the header, a green banner reads 'Ultra-fast Search Library and Server' and includes the Lucene and Solr logos. A dark blue banner below the green one states 'Apache Lucene and Solr set the standard for search and indexing performance'. The main content area is titled 'Welcome to Apache Lucene' and lists the project's components: Lucene Core, Solr, and PyLucene. To the right of the text, there are two 'DOWNLOAD' buttons for Apache Lucene 6.0.0 and Apache Solr 6.0.0. At the bottom right, a 'Projects' section lists 'Lucene Core (Java)'.

Apache Lucene - Welcome X

https://lucene.apache.org

Search with Apache Solr. @ select provider

Lucene™

CORE (JAVA) SOLR PYLUCENE

Ultra-fast Search Library and Server

Lucene Solr

Apache Lucene and Solr set the standard for search and indexing performance

Welcome to Apache Lucene

The Apache Lucene™ project develops open-source search software, including:

- [Lucene Core](#), our flagship sub-project, provides Java-based indexing and search technology, as well as spellchecking, hit highlighting and advanced analysis/tokenization capabilities.
- [Solr™](#) is a high performance search server built using Lucene Core, with XML/HTTP and JSON/Python/Ruby APIs, hit highlighting, faceted search, caching, replication, and a web admin interface.
- [PyLucene](#) is a Python port of the Core project.

DOWNLOAD

Apache Lucene 6.0.0

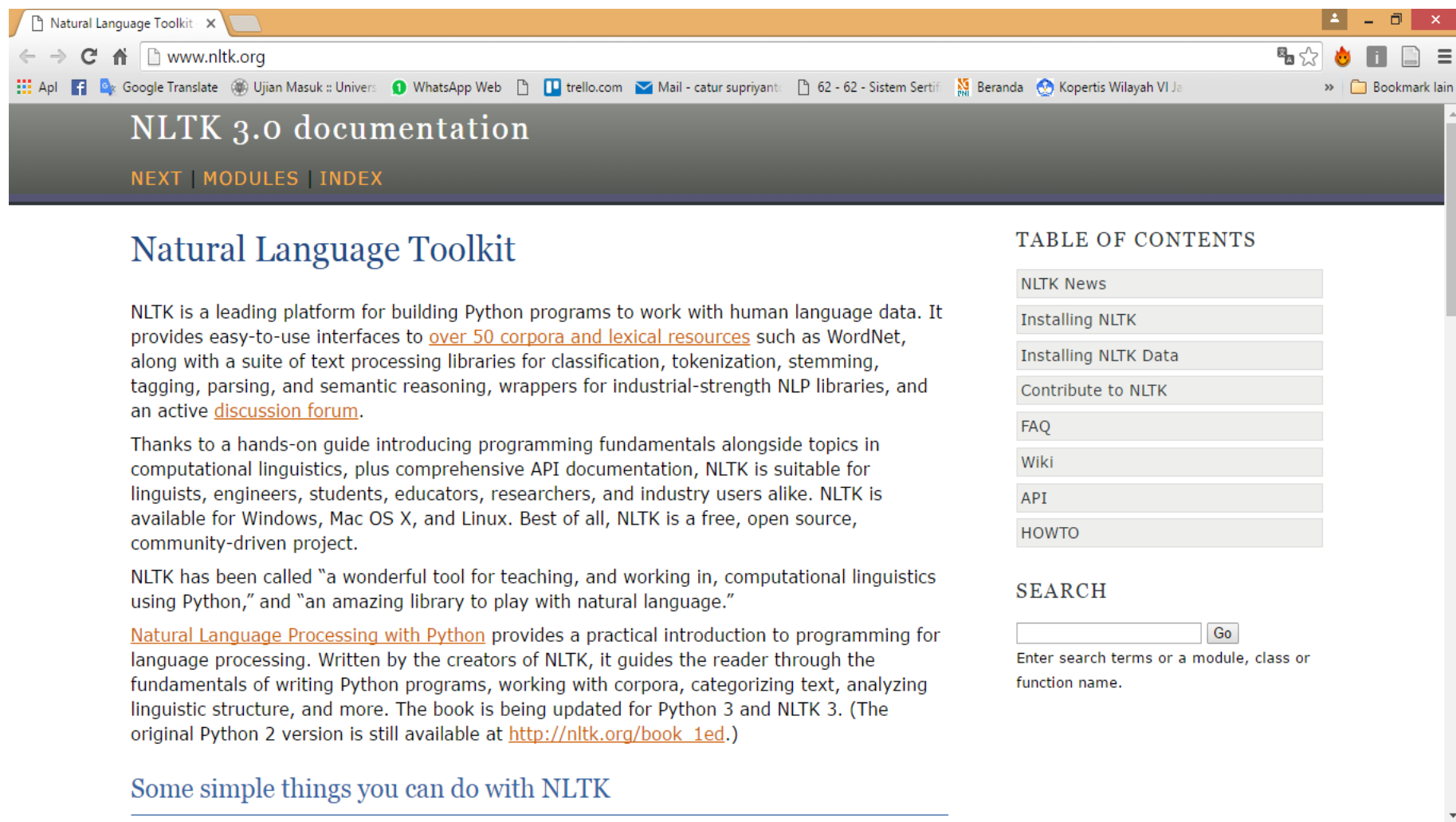
DOWNLOAD

Apache Solr 6.0.0

Projects

[Lucene Core \(Java\)](#)

NLTK



The screenshot shows a web browser window with the title "Natural Language Toolkit". The address bar shows "www.nltk.org". The browser's bookmark bar includes links to "Apl", "Google Translate", "Ujian Masuk :: Univers", "WhatsApp Web", "trello.com", "Mail - catur supriyant", "62 - 62 - Sistem Sertif", "Beranda", and "Kopertis Wilayah VI Je". The page content features a dark header with "NLTK 3.0 documentation" and navigation links "NEXT | MODULES | INDEX". The main heading is "Natural Language Toolkit". The text describes NLTK as a leading platform for building Python programs to work with human language data, providing interfaces to over 50 corpora and lexical resources, and a suite of text processing libraries. It also mentions a hands-on guide for programming fundamentals and API documentation. A "TABLE OF CONTENTS" sidebar lists links to "NLTK News", "Installing NLTK", "Installing NLTK Data", "Contribute to NLTK", "FAQ", "Wiki", "API", and "HOWTO". A "SEARCH" section includes a search box and a "Go" button. The footer contains the text "Some simple things you can do with NLTK".

Natural Language Toolkit

NLTK is a leading platform for building Python programs to work with human language data. It provides easy-to-use interfaces to [over 50 corpora and lexical resources](#) such as WordNet, along with a suite of text processing libraries for classification, tokenization, stemming, tagging, parsing, and semantic reasoning, wrappers for industrial-strength NLP libraries, and an active [discussion forum](#).

Thanks to a hands-on guide introducing programming fundamentals alongside topics in computational linguistics, plus comprehensive API documentation, NLTK is suitable for linguists, engineers, students, educators, researchers, and industry users alike. NLTK is available for Windows, Mac OS X, and Linux. Best of all, NLTK is a free, open source, community-driven project.

NLTK has been called "a wonderful tool for teaching, and working in, computational linguistics using Python," and "an amazing library to play with natural language."

[Natural Language Processing with Python](#) provides a practical introduction to programming for language processing. Written by the creators of NLTK, it guides the reader through the fundamentals of writing Python programs, working with corpora, categorizing text, analyzing linguistic structure, and more. The book is being updated for Python 3 and NLTK 3. (The original Python 2 version is still available at http://nltk.org/book_1ed.)

[Some simple things you can do with NLTK](#)

TABLE OF CONTENTS

- NLTK News
- Installing NLTK
- Installing NLTK Data
- Contribute to NLTK
- FAQ
- Wiki
- API
- HOWTO

SEARCH

Enter search terms or a module, class or function name.

Rapidminer

The screenshot displays the RapidMiner 5.3.015 interface. The main window shows a workflow diagram titled "Main Process" with the following components:

- Process Control** (37)
- Utility** (54)
- Repository Access** (6)
- Import** (27)
- Export** (18)
- Data Transformation** (115)
- Modeling** (118)
- Evaluation** (29)
- Text Processing** (48)

The workflow diagram includes the following nodes:

- Read Excel**: A node for reading data from an Excel file.
- Process Docu...**: A node for processing documents.
- Clustering**: A node for performing clustering analysis.
- Map Clusterin...**: A node for mapping cluster results.
- Performance**: A node for evaluating the performance of the model.

The right sidebar shows the **Parameters** tab for the selected **Process** node, with the following settings:

- logverbosity**: init
- logfile**: [empty]
- resultfile**: [empty]
- random seed**: 2001
- send mail**: never
- encoding**: SYSTEM

The bottom status bar shows **No problems found** and a **Log** tab.

Weka

[Project](#)

[Software](#)

[Book](#)

[Courses](#)

[Publications](#)

[People](#)

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Weka 3: Machine Learning Software in Java

Weka is a collection of machine learning algorithms for data mining tasks. It contains tools for data preparation, classification, regression, clustering, association rules mining, and visualization.

Found only on the islands of New Zealand, the Weka is a flightless bird with an inquisitive nature. The name is pronounced like **this**, and the bird sounds like **this**.

Weka is open source software issued under the **GNU General Public License**.

We have put together several **free online courses** that teach machine learning and data mining using Weka. The videos for the courses are available **on Youtube**.

Weka supports **deep learning**!

[Getting started](#)

[Further information](#)

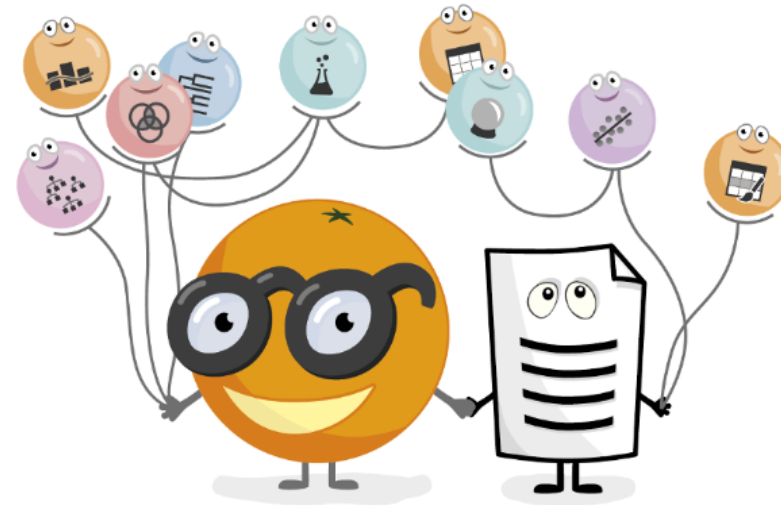
[Developers](#)

Orange

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[Doctoral Summer School](#)

Jul 25, 2019

[Orange at ISMB/ECCB 2019](#)

Text Clustering - Carrot2

The screenshot displays the Carrot2 search engine interface. The browser window title is "bean - Carrot2 Clustering". The address bar shows the URL: `search.carrot2.org/stable/search?source=web&view=folders&skin=fancy-compact&query=bean&results=100&algorithm=lingo&EToolsDocuments`. The search bar contains the query "bean".

On the left side, there is a "Folders" panel with a tree structure:

- All Topics (97)
 - Rowan Atkinson (12)
 - YouTube (9)
 - Bean Film (6)
 - Black Beans (5)
 - Garden (5)
 - Original Beans (5)
 - Plants (5)
 - Jelly Bean (4)
 - Bean Institute (3)
 - Coffee Bean Tea Leaf (3)
 - more | show all

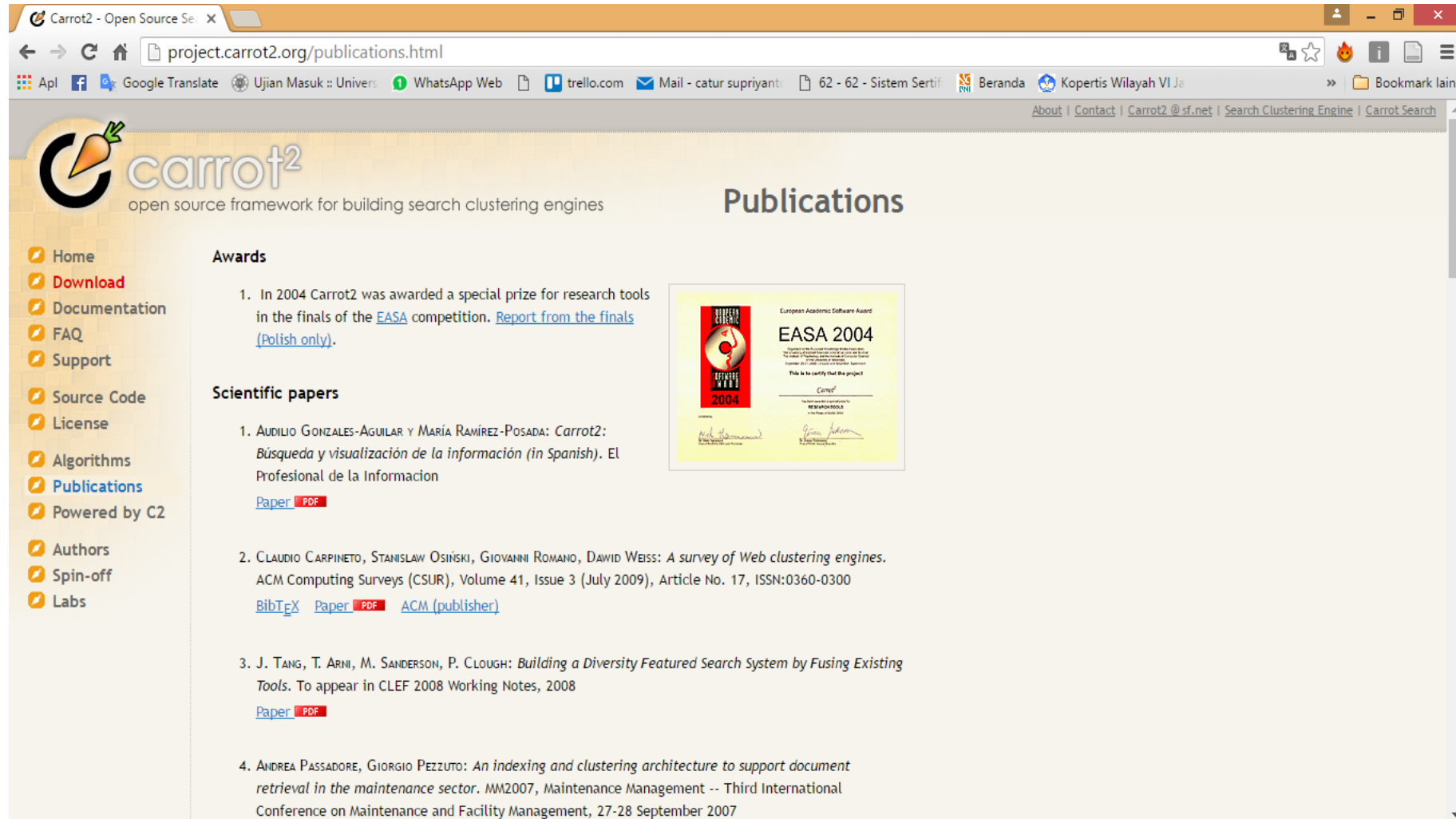
The main content area displays "Top 97 results of about 32100000 for bean". The results are numbered 1 through 8:

- 1: Bean - Wikipedia, the free encyclopedia**
Bean is a common name for large plant seeds of several genera of the family Fabaceae (alternately Leguminosae) which are used for human or animal food.
<https://en.wikipedia.org/wiki/Bean> [Ask, Bing, Goo, Google, Qwant, Wikipedia, Yahoo]
- 2: Bean (film) - Wikipedia, the free encyclopedia**
Bean is a 1997 feature film based on the television series Mr. Bean. It stars Rowan Atkinson in the title role and Peter MacNicol. It was directed by Mel Smith.
[https://en.wikipedia.org/wiki/Bean_\(film\)](https://en.wikipedia.org/wiki/Bean_(film)) [Goo, Google]
- 3: L.L.Bean: FREE Shipping. 100% Satisfaction Guaranteed.**
Shop L.L.Bean for Free Shipping on clothing, shoes, outdoor gear and more, all backed by our 100% satisfaction guarantee.
<http://www.lbean.com/> [Ask, Bing, Goo, Google, Qwant, Yahoo]
- 4: Bean (1997) - IMDb**
With Rowan Atkinson, Peter MacNicol, John Mills, Pamela Reed. The bumbling Mr. Bean travels to America when he is given the responsibility of bringing a ...
<http://www.imdb.com/title/tt0118689/> [Ask, Bing, Goo, Google, Qwant, Yahoo]
- 5: Bean: A Word Processor for OS X**
Bean is a small, easy-to-use word processor that is designed to make writing convenient, efficient and comfortable. Bean is lean, fast and uncluttered. It starts up ...
<http://www.bean-osx.com/> [Ask, Goo, Google]
- 6: Mr Bean: Home**
Official site. Features an overview, character and episode guides, clips, a FAQ and factfile.
<http://www.mrbean.com/> [Ask, Goo, Google, Qwant]
- 7: Mr. Bean - YouTube**
Welcome to the Official Mr Bean channel. The first episode of the original Mr Bean series starring Rowan Atkinson was first broadcast on 1st January 1990. Si ...
<https://www.youtube.com/user/MrBean/videos> [Ask, Goo, Google]
- 8: The Bean**

The footer of the page contains the following information:

Query: bean -- Source: Web (97 results, 3762 ms) -- Clusterer: Lingo (81 ms) v3.11.0 | build 163 | 2015-10-19 20:31 © 2002-2016 Stanislaw Osinski, David Weiss

Carrot2



The screenshot shows a web browser window with the address bar displaying `project.carrot2.org/publications.html`. The website has a light beige background with a grid pattern. The Carrot2 logo, featuring a stylized carrot with a green leaf, is in the top left corner, with the text "carrot²" and "open source framework for building search clustering engines" below it. A navigation menu on the left lists various links: Home, Download, Documentation, FAQ, Support, Source Code, License, Algorithms, Publications, Powered by C2, Authors, Spin-off, and Labs. The main content area is titled "Publications" and is divided into two sections: "Awards" and "Scientific papers". The "Awards" section mentions a special prize for research tools in the EASA competition in 2004, with a link to a report. The "Scientific papers" section lists four papers, each with a title, authors, and a link to the paper or PDF. A European Academic Software Award (EASA) 2004 certificate is displayed on the right side of the page.

Carrot2 - Open Source Search Engine

project.carrot2.org/publications.html

Apl | Google Translate | Ujian Masuk :: Univers | WhatsApp Web | trello.com | Mail - catur supriyanti | 62 - 62 - Sistem Sertifi | Beranda | Kopertis Wilayah VI Ja | Bookmark lain

About | Contact | Carrot2 @ sf.net | Search Clustering Engine | Carrot Search

carrot²
open source framework for building search clustering engines

Publications

Awards

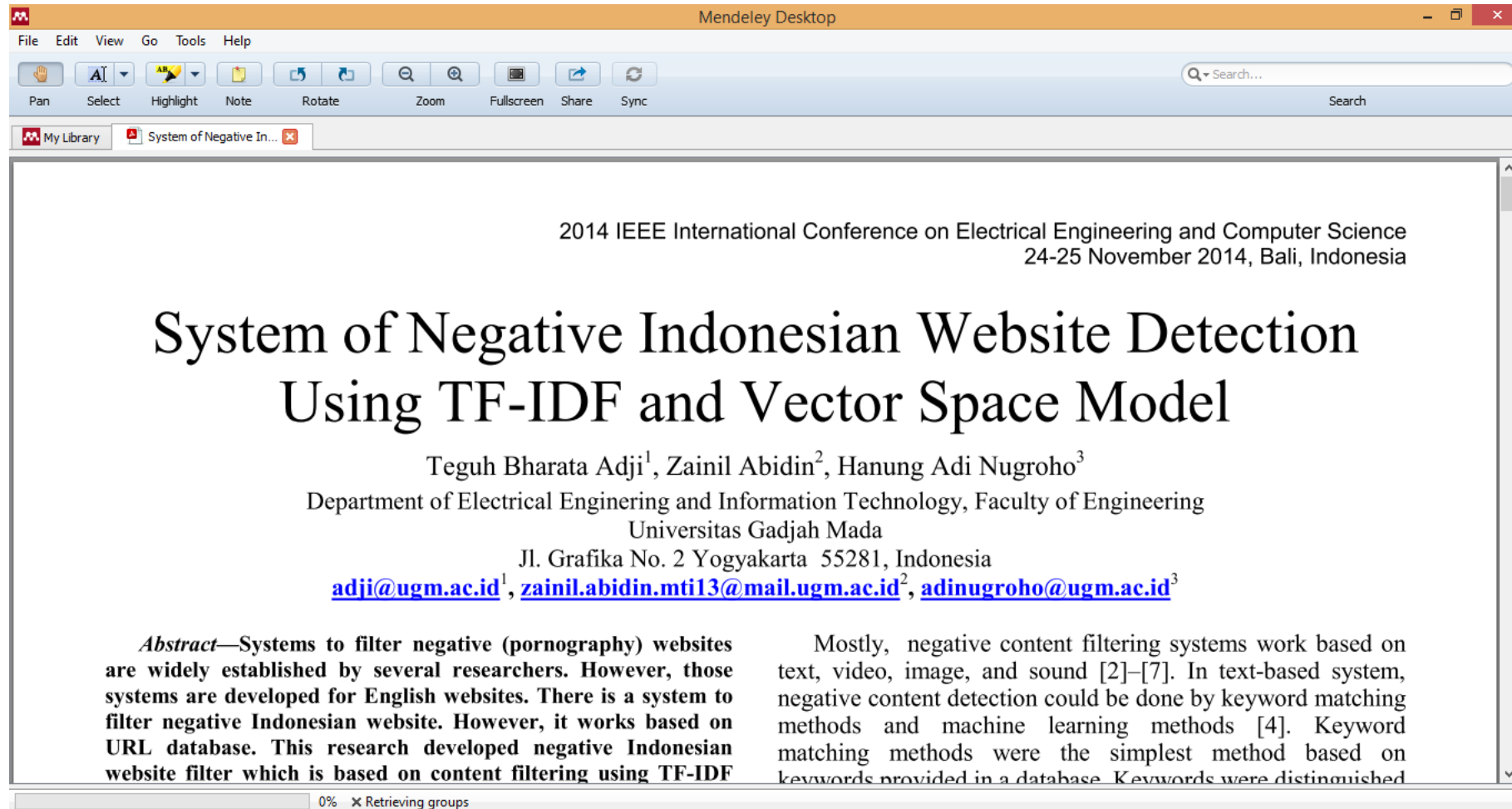
1. In 2004 Carrot2 was awarded a special prize for research tools in the finals of the [EASA](#) competition. [Report from the finals \(Polish only\)](#).

Scientific papers

1. AUDILIO GONZALES-AGUILAR Y MARÍA RAMÍREZ-POSADA: *Carrot2: Búsqueda y visualización de la información (in Spanish)*. El Profesional de la Información
[Paper](#) [PDF](#)
2. CLAUDIO CARPINETO, STANISLAW OSIŃSKI, GIOVANNI ROMANO, DAWID WEISS: *A survey of Web clustering engines*. ACM Computing Surveys (CSUR), Volume 41, Issue 3 (July 2009), Article No. 17, ISSN:0360-0300
[BibTeX](#) [Paper](#) [PDF](#) [ACM \(publisher\)](#)
3. J. TANG, T. ARNI, M. SANDERSON, P. CLOUGH: *Building a Diversity Featured Search System by Fusing Existing Tools*. To appear in CLEF 2008 Working Notes, 2008
[Paper](#) [PDF](#)
4. ANDREA PASSADORE, GIORGIO PEZZUTO: *An indexing and clustering architecture to support document retrieval in the maintenance sector*. MM2007, Maintenance Management -- Third International Conference on Maintenance and Facility Management, 27-28 September 2007

European Academic Software Award
EASA 2004
This is to certify that the project
Carrot2
has been awarded the
European Academic Software Award
for the year 2004

Porn Detection



The screenshot shows a Mendeley Desktop application window. The title bar reads 'Mendeley Desktop'. The menu bar includes 'File', 'Edit', 'View', 'Go', 'Tools', and 'Help'. The toolbar contains icons for 'Pan', 'Select', 'Highlight', 'Note', 'Rotate', 'Zoom', 'Fullscreen', 'Share', and 'Sync'. A search bar is located on the right side of the toolbar. The main content area displays a PDF document with the following text:

2014 IEEE International Conference on Electrical Engineering and Computer Science
24-25 November 2014, Bali, Indonesia

System of Negative Indonesian Website Detection Using TF-IDF and Vector Space Model

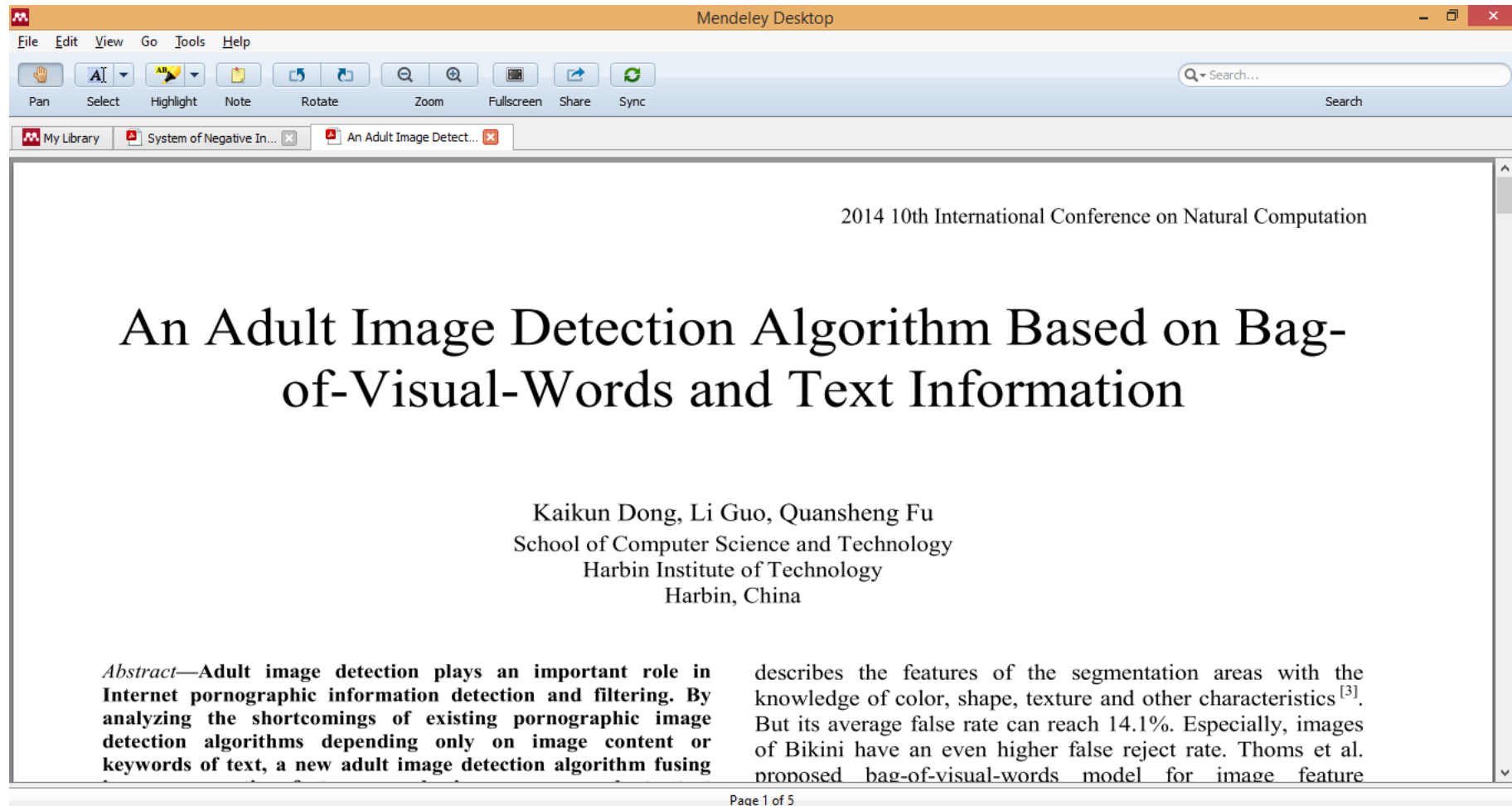
Teguh Bharata Adji¹, Zainil Abidin², Hanung Adi Nugroho³
Department of Electrical Engineering and Information Technology, Faculty of Engineering
Universitas Gadjah Mada
Jl. Grafika No. 2 Yogyakarta 55281, Indonesia
adji@ugm.ac.id¹, zainil.abidin.mti13@mail.ugm.ac.id², adinugroho@ugm.ac.id³

Abstract—Systems to filter negative (pornography) websites are widely established by several researchers. However, those systems are developed for English websites. There is a system to filter negative Indonesian website. However, it works based on URL database. This research developed negative Indonesian website filter which is based on content filtering using TF-IDF

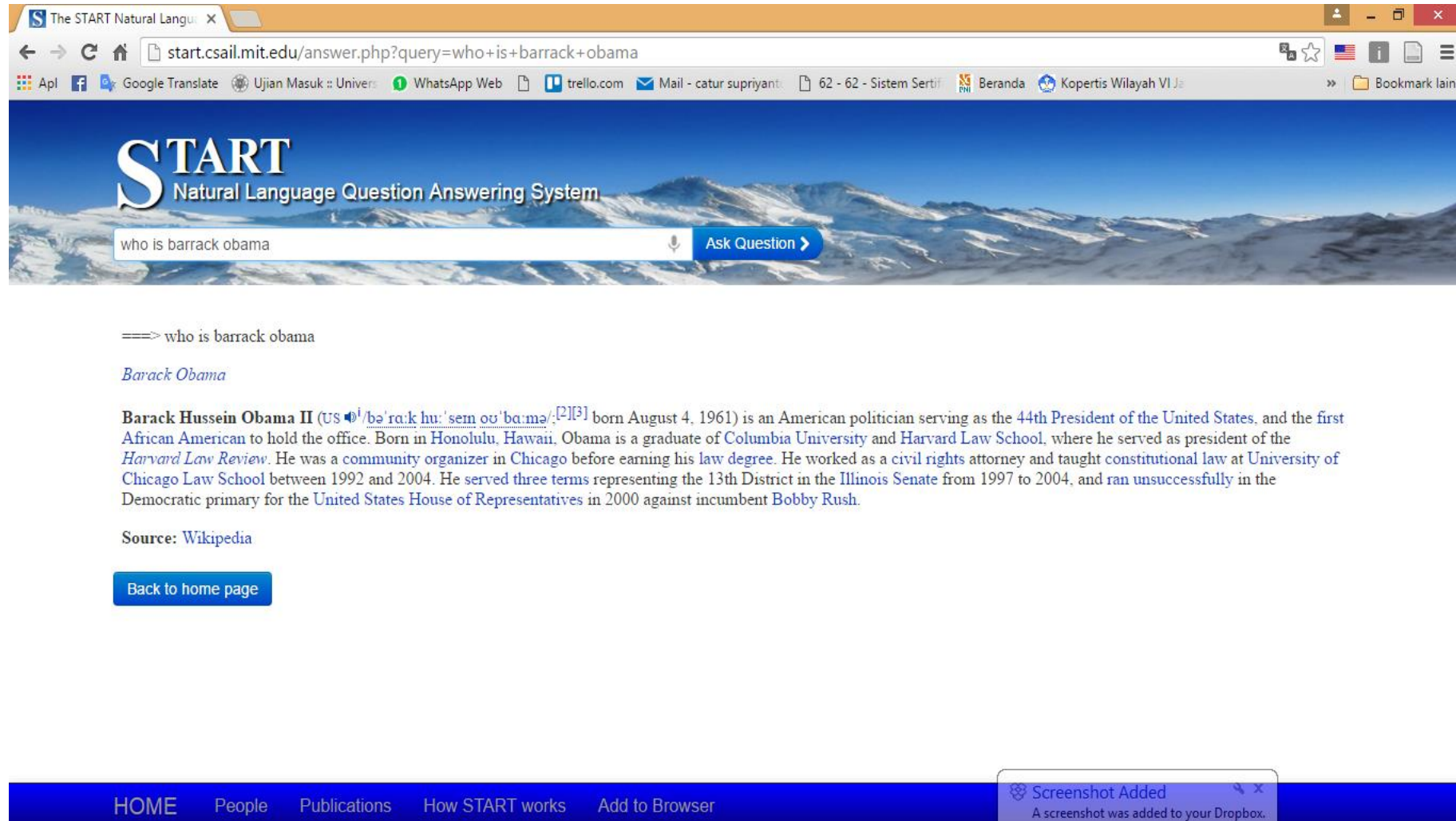
Mostly, negative content filtering systems work based on text, video, image, and sound [2]–[7]. In text-based system, negative content detection could be done by keyword matching methods and machine learning methods [4]. Keyword matching methods were the simplest method based on keywords provided in a database. Keywords were distinguished

0% Retrieving groups

Text and Image Based Porn Detection



Question Answering System



The screenshot shows a web browser window with the URL `start.csail.mit.edu/answer.php?query=who+is+barrack+obama`. The page features a header with the "START Natural Language Question Answering System" logo and a search bar containing the query "who is barrack obama". Below the search bar, the system displays the query "====> who is barrack obama" and the answer "Barack Obama". The answer is followed by a detailed paragraph about Barack Hussein Obama II, including his birth date, education, and political career. The source is cited as "Wikipedia". A "Back to home page" button is located at the bottom left. The footer contains navigation links: "HOME", "People", "Publications", "How START works", and "Add to Browser". A notification at the bottom right states "Screenshot Added" and "A screenshot was added to your Dropbox."

The START Natural Language Question Answering System

who is barrack obama

====> who is barrack obama

Barack Obama

Barack Hussein Obama II (US ⁱ/bəˈrɑːk huːˈsem ʊˈbɑːmə/^[2]^[3] born August 4, 1961) is an American politician serving as the 44th President of the United States, and the first African American to hold the office. Born in Honolulu, Hawaii, Obama is a graduate of Columbia University and Harvard Law School, where he served as president of the *Harvard Law Review*. He was a community organizer in Chicago before earning his law degree. He worked as a civil rights attorney and taught constitutional law at University of Chicago Law School between 1992 and 2004. He served three terms representing the 13th District in the Illinois Senate from 1997 to 2004, and ran unsuccessfully in the Democratic primary for the United States House of Representatives in 2000 against incumbent Bobby Rush.

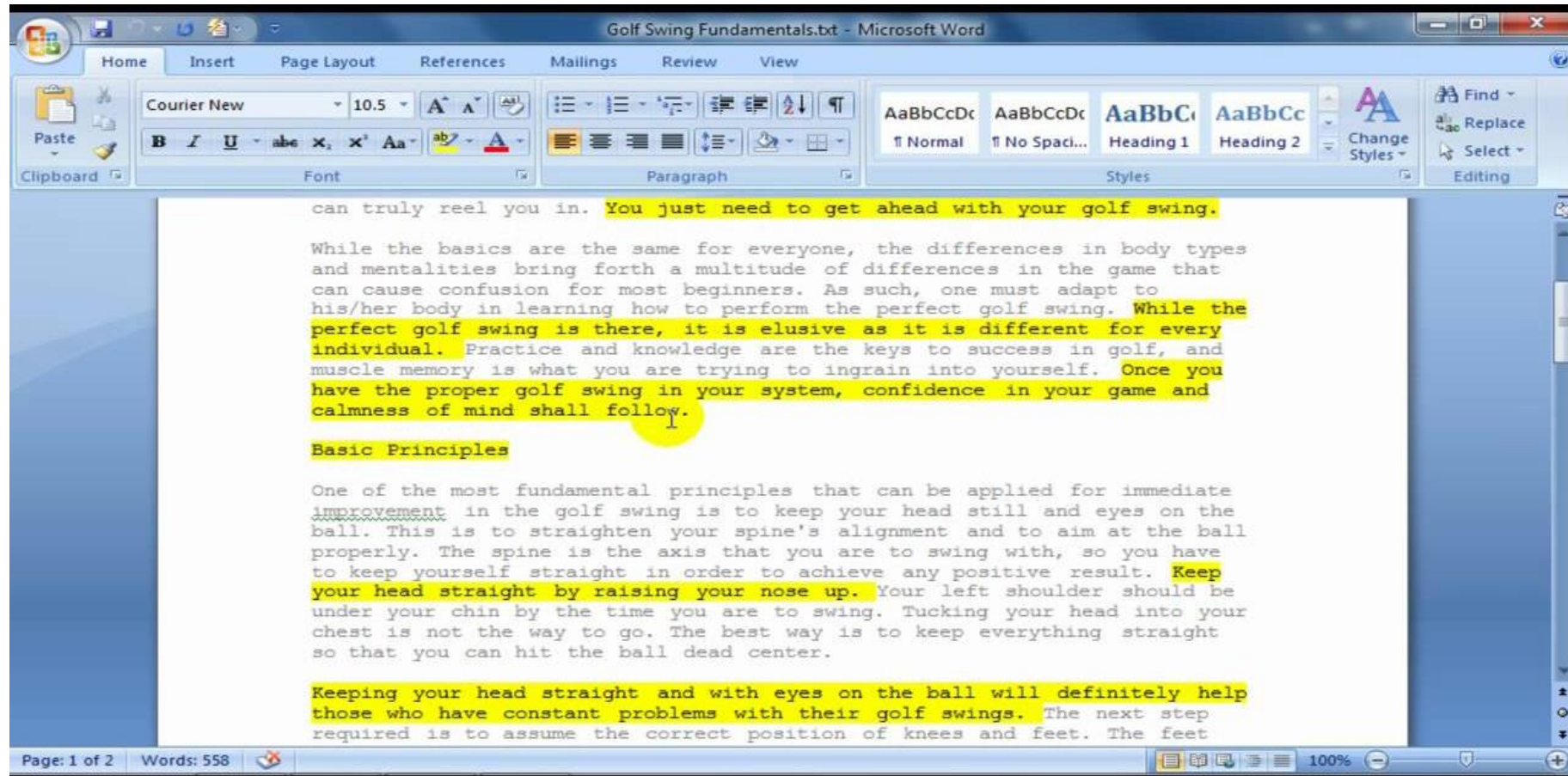
Source: Wikipedia

[Back to home page](#)

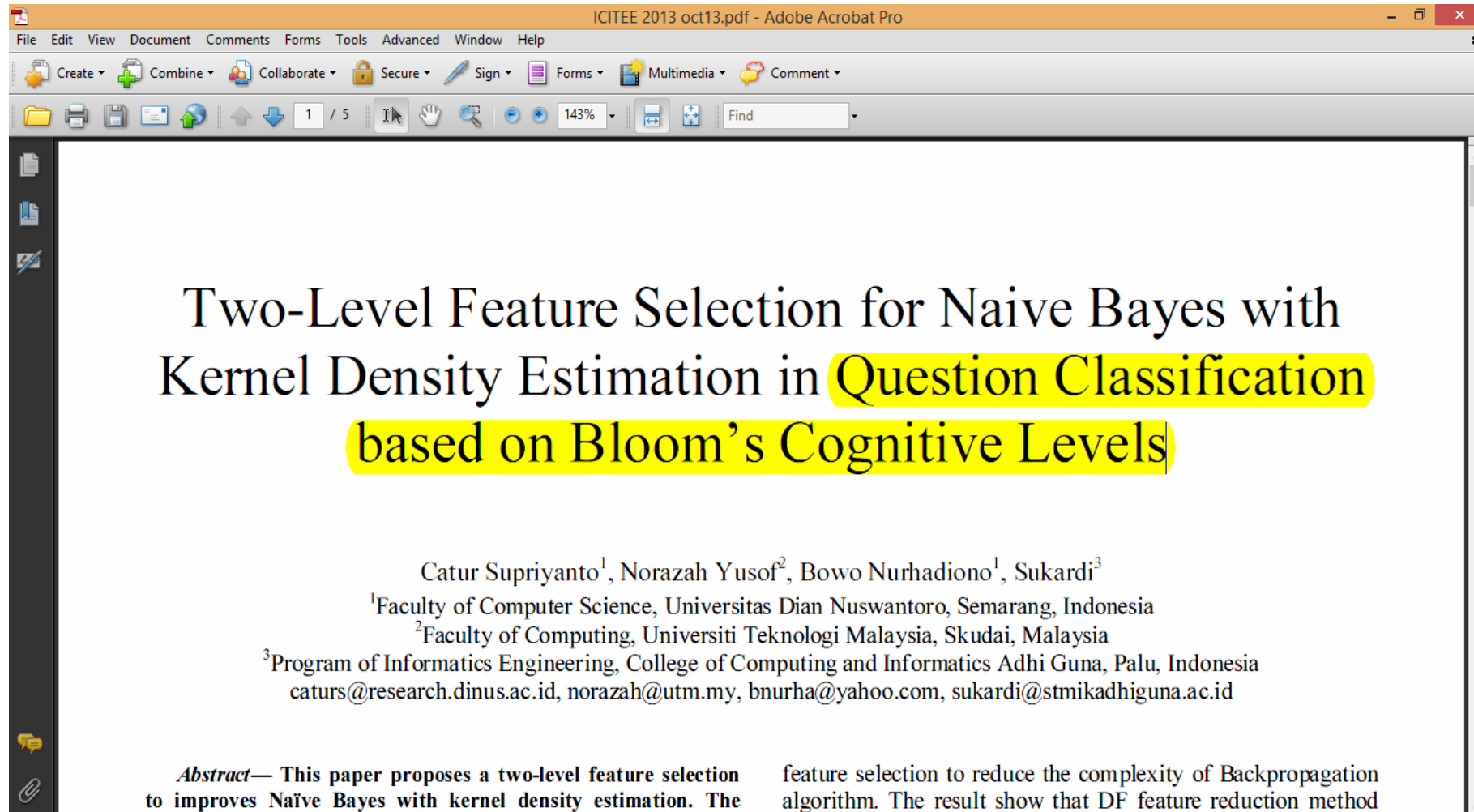
HOME People Publications How START works Add to Browser

Screenshot Added
A screenshot was added to your Dropbox.

Text Summarization



Question Classification



Bloom Taxonomy

domain of bloom's taxonomy.

TABLE I KEYWORDS USED IN KOGNITIVE DOMAIN [15]

Category	Keywords
Knowledge	Defines, describes, identifies, knows, labels, lists, matches, names, outlines, recalls, recognizes, reproduces, selects, states.
Comprehension	Comprehends converts, defends, distinguishes, estimates, explains, extends, generalizes, gives examples, infers, interprets, paraphrases, predicts, rewrites, summarizes, translates.
Application	Applies, changes, computes, constructs, demonstrates, discovers, manipulates, modifies, operates, predicts, prepares, produces, relates, shows, solves, uses.
Analysis	Analyzes, breaks down, compares, contrasts, diagrams, deconstructs, differentiates, discriminates, distinguishes, identifies, illustrates, infers, outlines, relates, selects, separates.
Synthesis	Categorizes, combines, compiles, composes, creates, devises, designs, explains, generates, modifies, organizes, plans, rearranges, reconstructs, relates, reorganizes, revises, rewrites, summarizes, tells, writes.
Evaluation	Appraises, compares, concludes, contrasts, criticizes, critiques, defends, describes, discriminates, evaluates, explains, interprets, justifies, relates, summarizes, supports.

B. Naïve Bayes Classifier

Where T_c is the number of times the word w_i that occur in class C , M is the number of words in category C , N is The size of the vocabulary table, λ is the positive constant, usually 1, or 0.5 to avoid zero probability.

C. Naïve Bayes with Kernel Density Estimation

Kernel Density Estimation (KDE) can manipulate quantitative attributes for naïve-Bayes [17]. To deal with the quantitative data, naïve Bayes use normal Gaussian distribution.

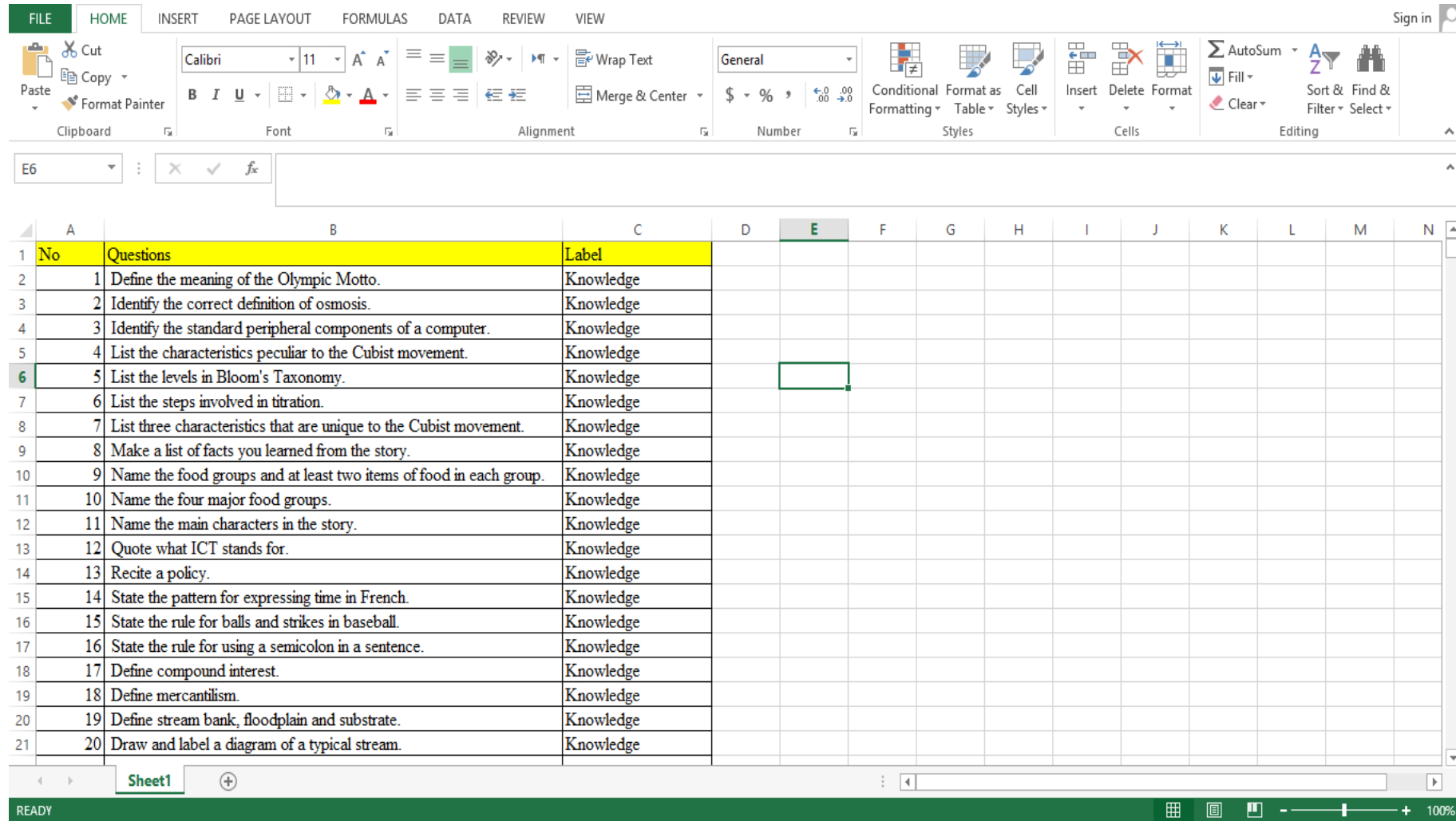
$$f = g(x, \mu_i, \sigma_c) = \frac{1}{\sqrt{2\pi}\sigma_c} e^{-\frac{(x-\mu_i)^2}{2\sigma_c^2}} \quad (5)$$
$$P(D = d | C = c) = \frac{1}{n} \sum_i g(x, \mu_i, \sigma_c) \quad (6)$$

Where i is the range of training data for the attribute x in class C , $\mu_i = x_i$ and $\sigma_c = \frac{1}{\sqrt{n_c}}$, n_c is the number of document in class C .

D. Filter Based Feature Selection

Filter based feature selection evaluates the features by

Question Classification Dataset



Sentiment Analysis

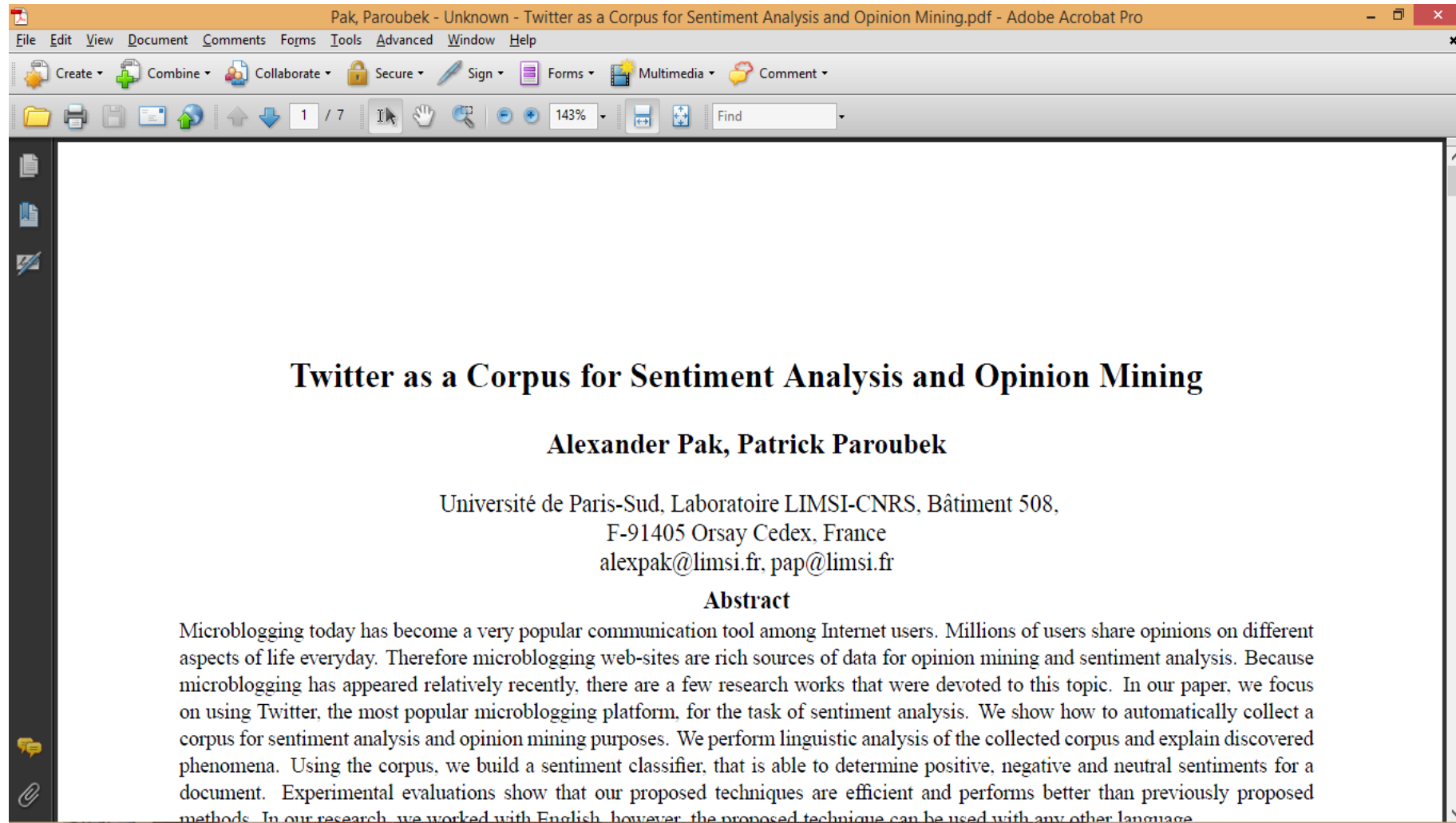
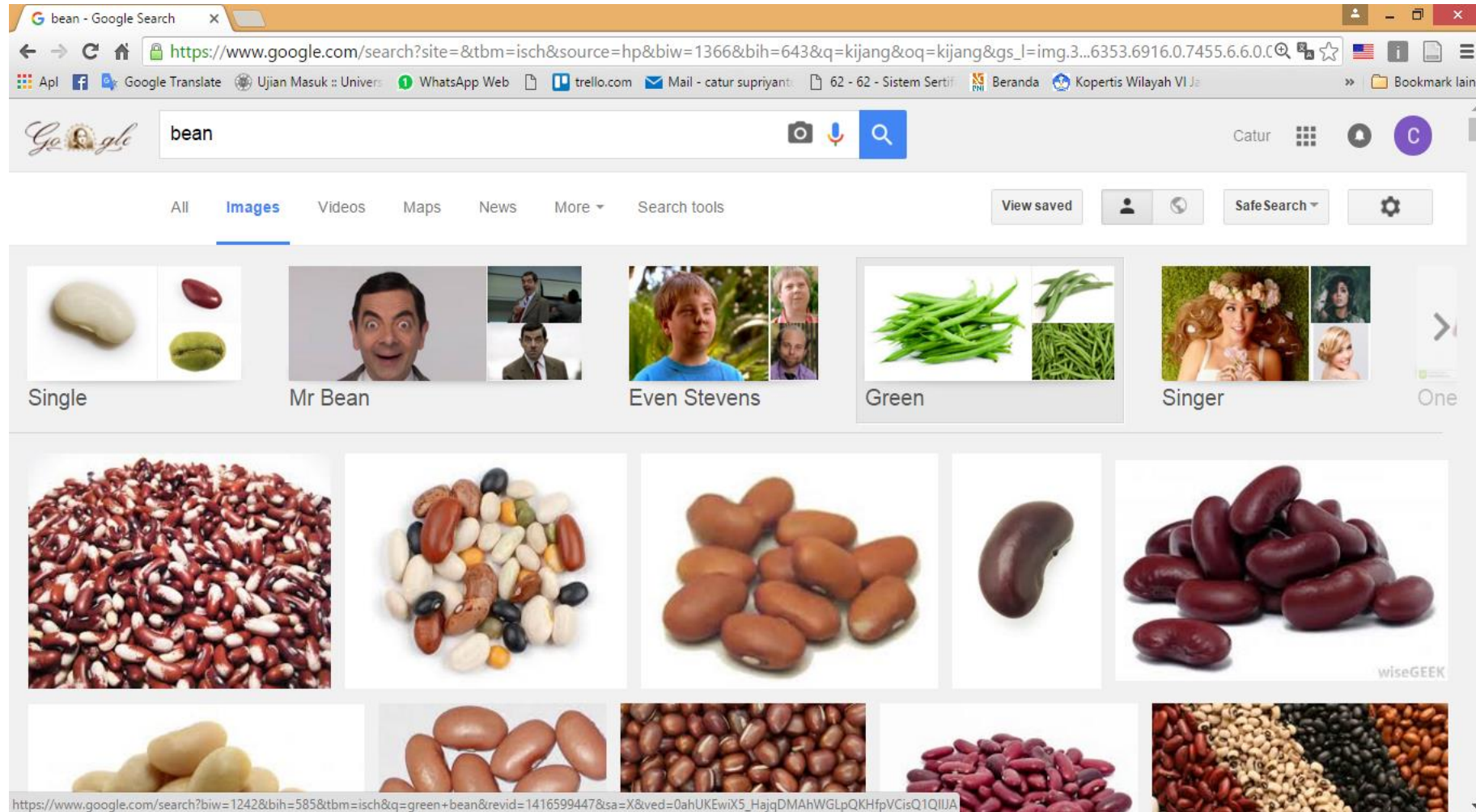


Image Clustering



Machine Learning for Text Mining

- Text mining typically apply machine learning techniques, such as **supervised learning** and **unsupervised learning**.
- Supervised learning is referred to **classification**.
- Unsupervised learning is referred to **clustering**.

Supervised Learning (SL)

Label:

1. Sport
2. Economy
3. Politic
4. Health

Labeled
Document
Training

SL Algorithm

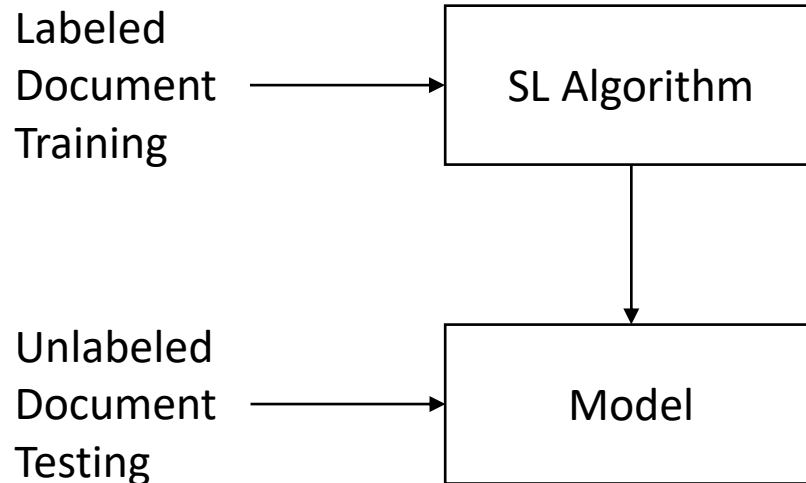
Algorithms:

1. Neural Network
2. Naïve Bayes
3. K-Nearest Neighbor
4. Support Vector Machine
5. Decision Tree

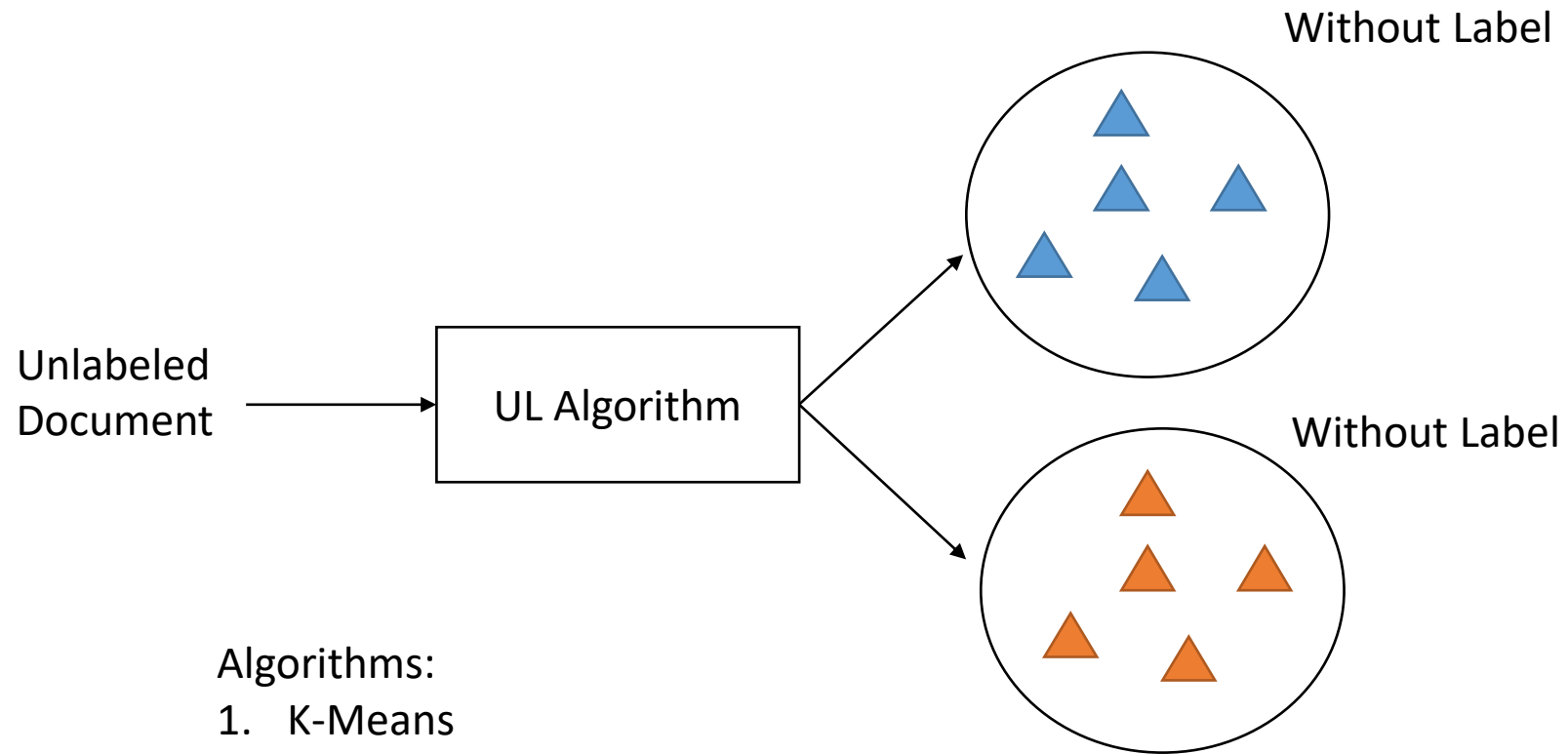
Unlabeled
Document
Testing

Model

Labeled
Document
Testing



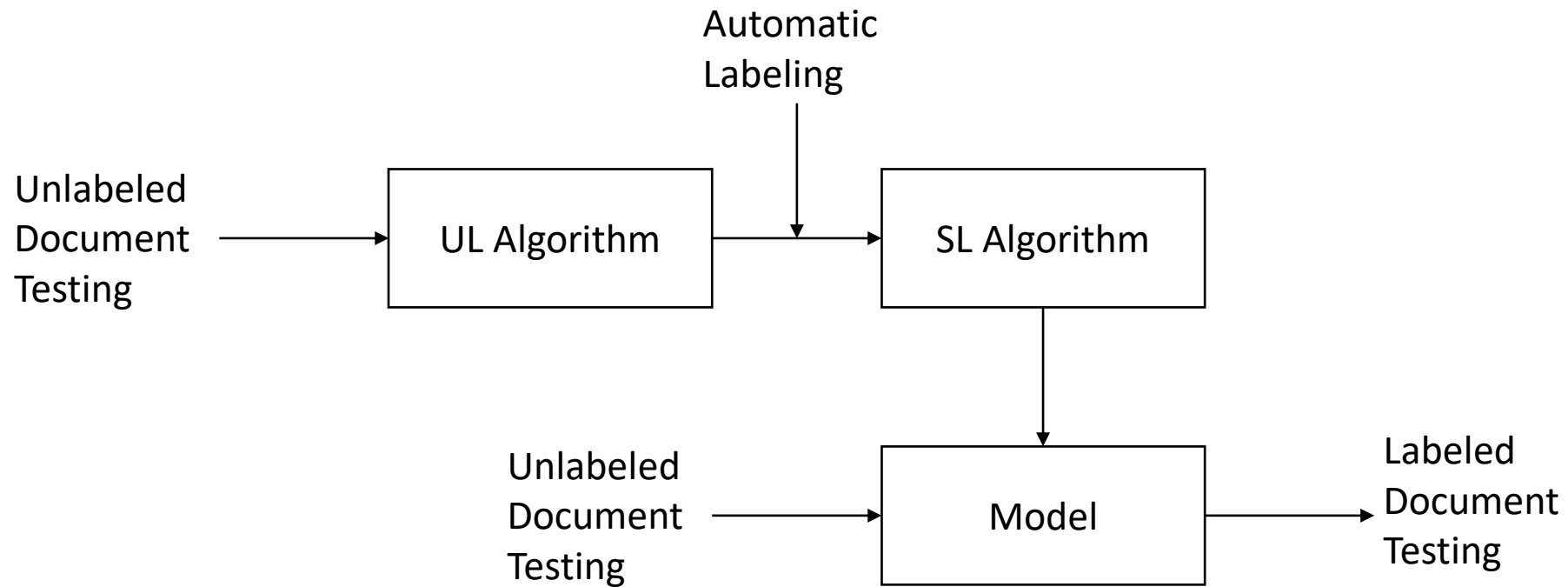
Unsupervised Learning (UL)



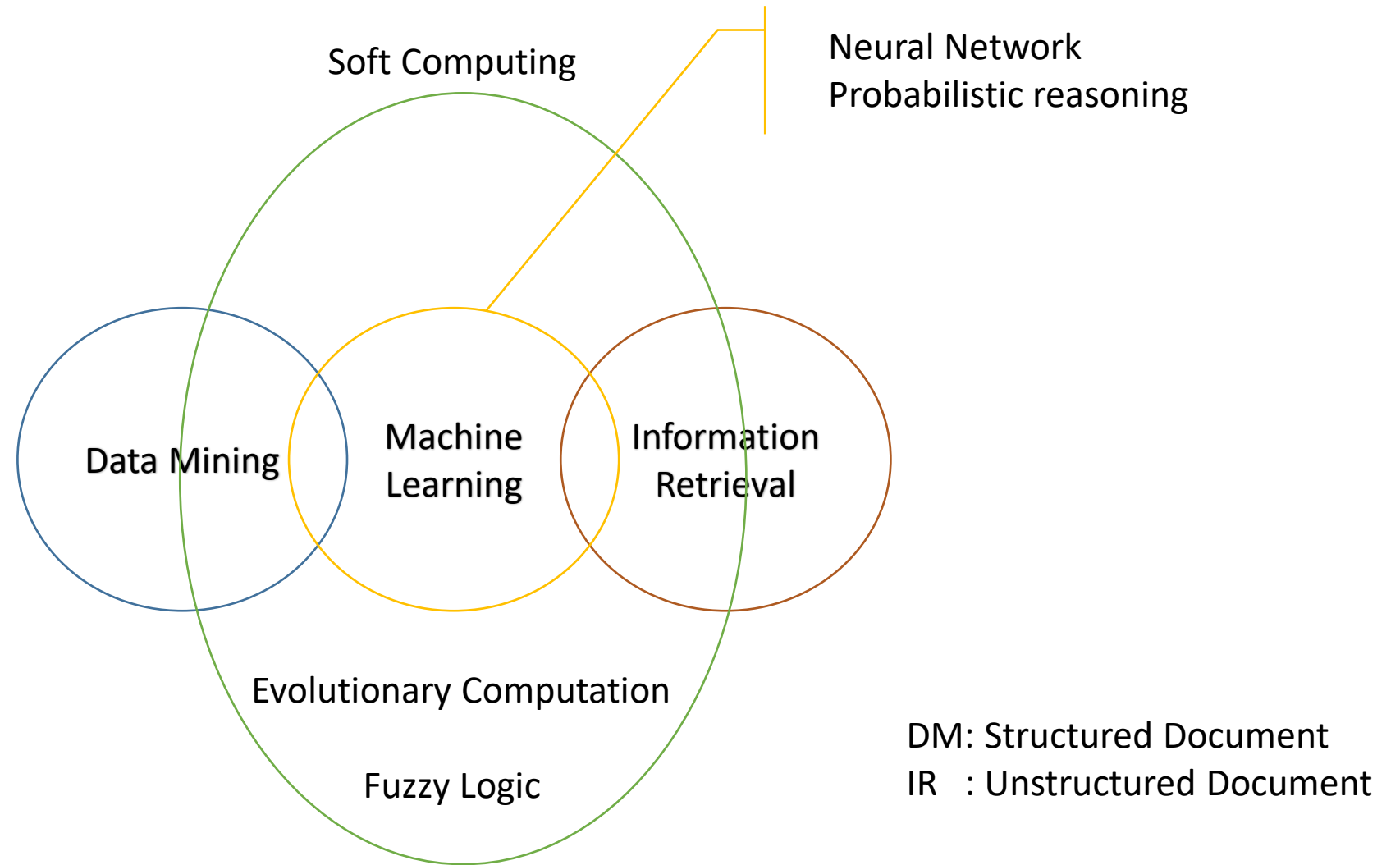
Algorithms:

1. K-Means
2. K-Medoid
3. Hierarchical Agglomerative Clustering (HAC)
4. Fuzzy C-Means (FCM)
5. Self Organizing Map (SOM)

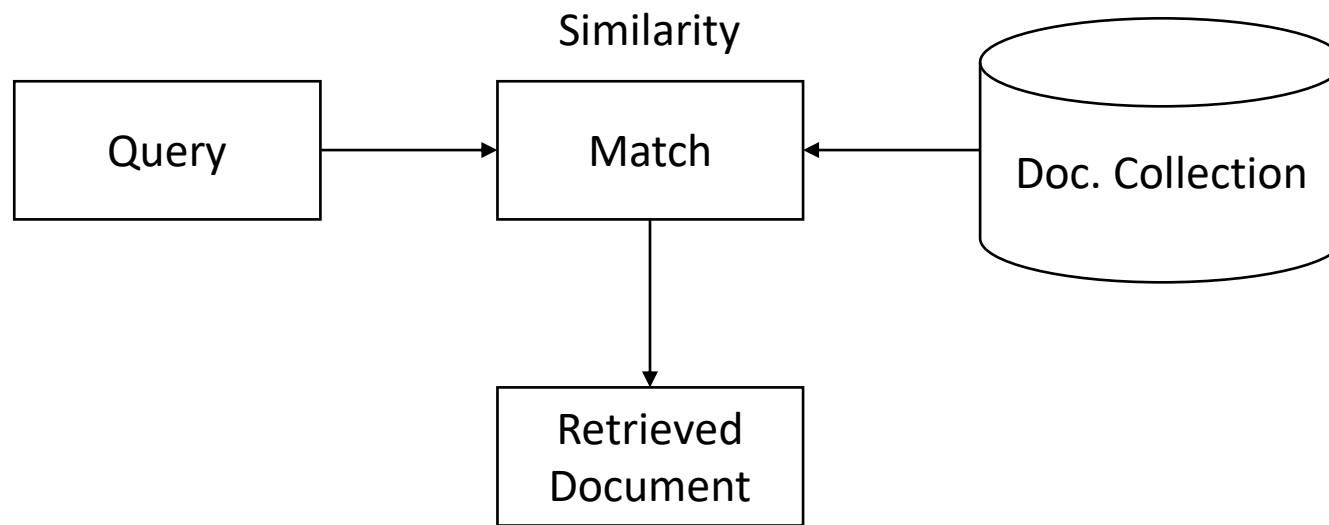
Supervised Learning (SL) and Unsupervised Learning (UL)



DM vs IR



Searching



Cosines Similarity

$$\text{Sim}(x, y) = \frac{\sum_{i=1}^N x_i \times y_i}{\sqrt{\sum_{i=1}^N (x_i)^2} \times \sqrt{\sum_{i=1}^N (y_i)^2}}$$

